MATH50003 Numerical Analysis

V.1 Fourier Expansions

Dr Sheehan Olver

Chapter V

Numerical Fourier series

1. Fourier Expansions and approximating Fourier coefficients
2. Discrete Fourier Transforms and interpolation
3. Convolutions via the DFT

V.1.1 Basics of Fourier series

Expanding functions in trigonometric polynomials

Definition 29 (Fourier)**.** A function f has a Fourier expansion if

$$f(\theta) = \sum_{k=-\infty}^{\infty} \hat{f}_k \mathrm{e}^{\mathrm{i}k\theta}$$

where

$$\hat{f}_k := \langle \mathrm{e}^{\mathrm{i}k\theta}, f \rangle = \frac{1}{2\pi} \int_0^{2\pi} \mathrm{e}^{-\mathrm{i}k\theta} f(\theta) \mathrm{d}\theta$$

Definition 30 (Fourier-Taylor)**.** A function f has a Fourier–Taylor expansion if

$$f(\theta) = \sum_{k=0}^{\infty} \hat{f}_k \mathrm{e}^{\mathrm{i}k\theta} = \sum_{k=0}^{\infty} \hat{f}_k z^k$$

where $\hat{f}_k := \langle \mathrm{e}^{\mathrm{i}k\theta}, f \rangle$, and $z = \mathrm{e}^{\mathrm{i}\theta}$.

Definition 31 (absolute convergent)**.** We write $\hat{\boldsymbol{f}} \in \ell^1$ if it is absolutely convergent, or in otherwords, the 1-norm of $\hat{\boldsymbol{f}}$ is bounded:

$$\|\hat{\boldsymbol{f}}\|_1 := \sum_{k=-\infty}^{\infty} |\hat{f}_k| < \infty.$$

Theorem 10 (Fourier series equivalence)**.** *If $f, g : \mathbb{T} \to \mathbb{C}$ are periodic and continuous and $\hat{f}_k = \hat{g}_k$ for all $k \in \mathbb{Z}$ then $f = g$.*

Proof See Körner 2022 (Theorem 2.4). ■

Theorem 11 (Absolute converging Fourier series)**.** *If* $\hat{\boldsymbol{f}} \in \ell^1$ *then*

$$f(\theta) = \sum_{k=-\infty}^{\infty} \hat{f}_k \mathrm{e}^{\mathrm{i}k\theta},$$

which converges uniformly.

Lemma 7 (differentiability and absolutely convergence)**.** *If $f : \mathbb{T} \to \mathbb{C}$ and f' are periodic and f'' is uniformly bounded, then $\hat{\boldsymbol{f}} \in \ell^1$.*

V.1.2 Trapezium rule and discrete Fourier coefficients

Using the Trapezium rule to approximate coefficients has nice structure

Definition 32 (Periodic Trapezium Rule)**.**

Lemma 8 (Discrete orthogonality)**.** *We have:*

$$\sum_{j=0}^{n-1} \mathrm{e}^{\mathrm{i}k\theta_j} = \begin{cases} n & k = \ldots, -2n, -n, 0, n, 2n, \ldots \\ 0 & \textit{otherwise} \end{cases}$$

V.1.3 Convergence of Approximate Fourier coefficients

Using Trapezium rule leads to a convergent approximation

Definition 33 (Discrete Fourier coefficients)**.** Define the Trapezium rule approximation to the Fourier coefficients by:

$$\hat{f}_k^n := \Sigma_n[\mathrm{e}^{-ik\theta} f(\theta)] = \frac{1}{n}\sum_{j=0}^{n-1} \mathrm{e}^{-ik\theta_j} f(\theta_j)$$

Theorem 12 (discrete Fourier coefficients)**.** *If* $\hat{\boldsymbol{f}} \in \ell^1$ *(absolutely convergent Fourier coefficients) then*

$$\hat{f}_k^n = \cdots + \hat{f}_{k-2n} + \hat{f}_{k-n} + \hat{f}_k + \hat{f}_{k+n} + \hat{f}_{k+2n} + \cdots$$

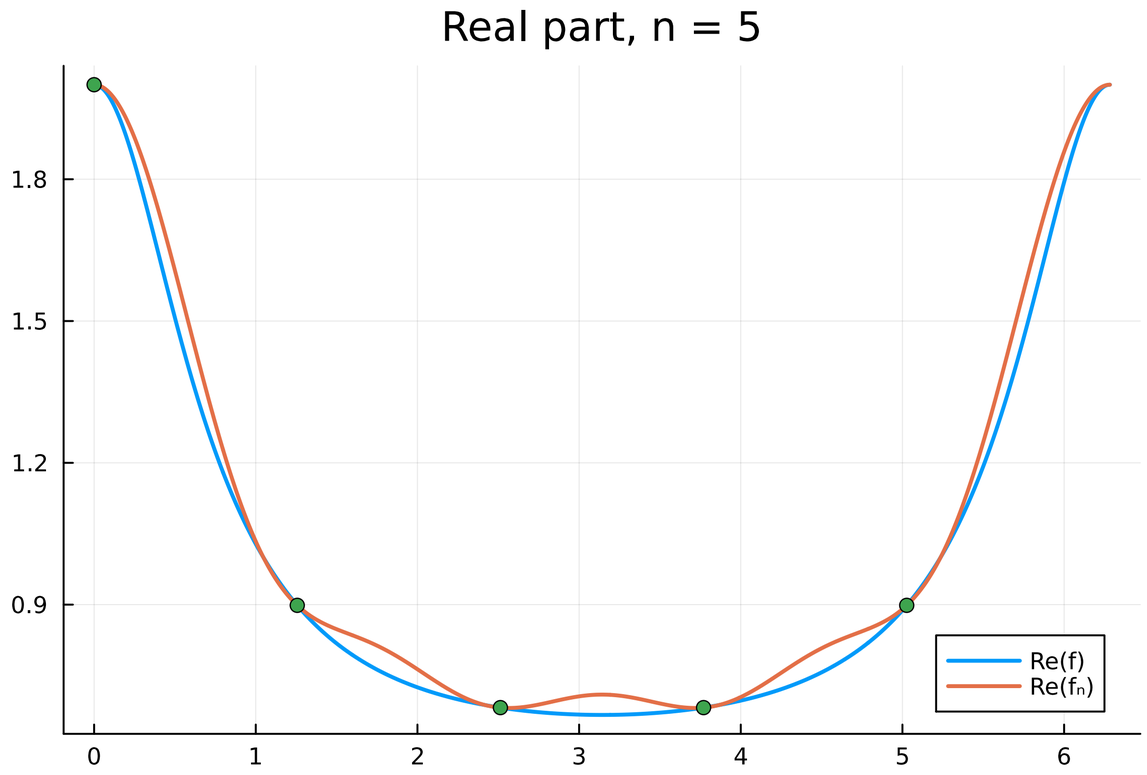
Example 20 (Taylor coefficients via Geometric series)**.**

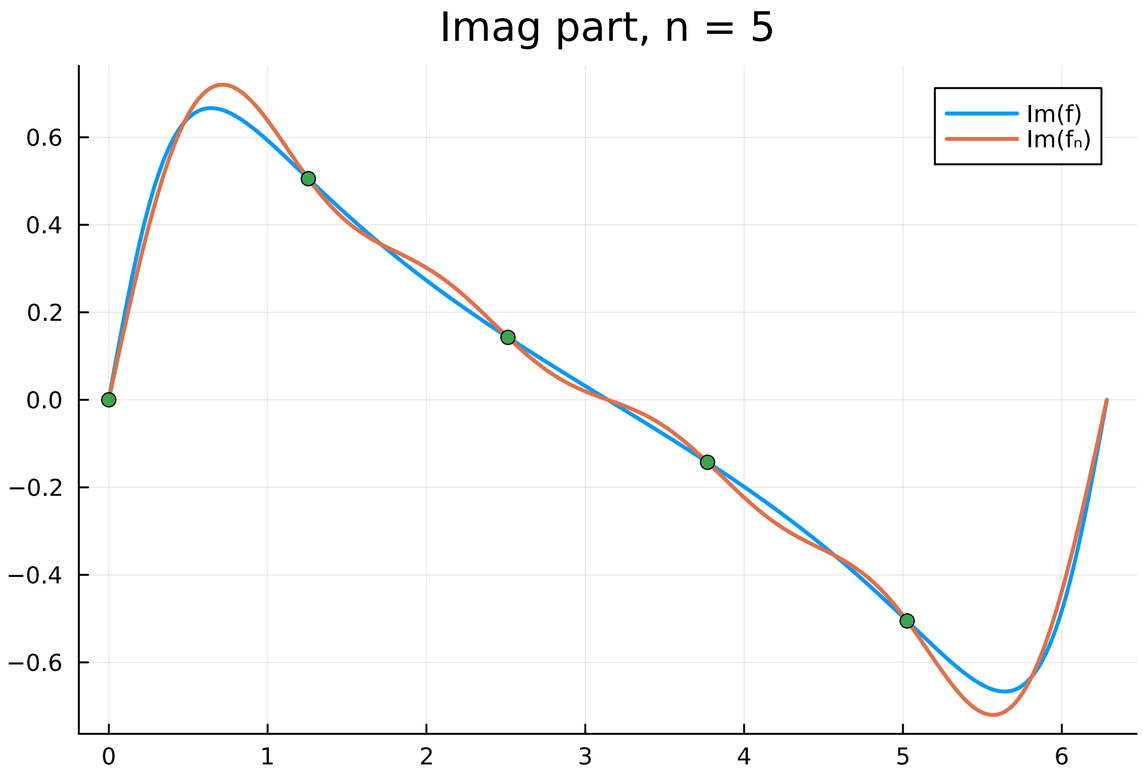
Theorem 13 (Approximate Fourier-Taylor expansions converge)**.** *If* $0 = \hat{f}_{-1} = \hat{f}_{-2} = \cdots$ *and* $\hat{\boldsymbol{f}}$ *is absolutely convergent then*

$$f_n(\theta) = \sum_{k=0}^{n-1} \hat{f}_k^n \mathrm{e}^{\mathrm{i}k\theta}$$

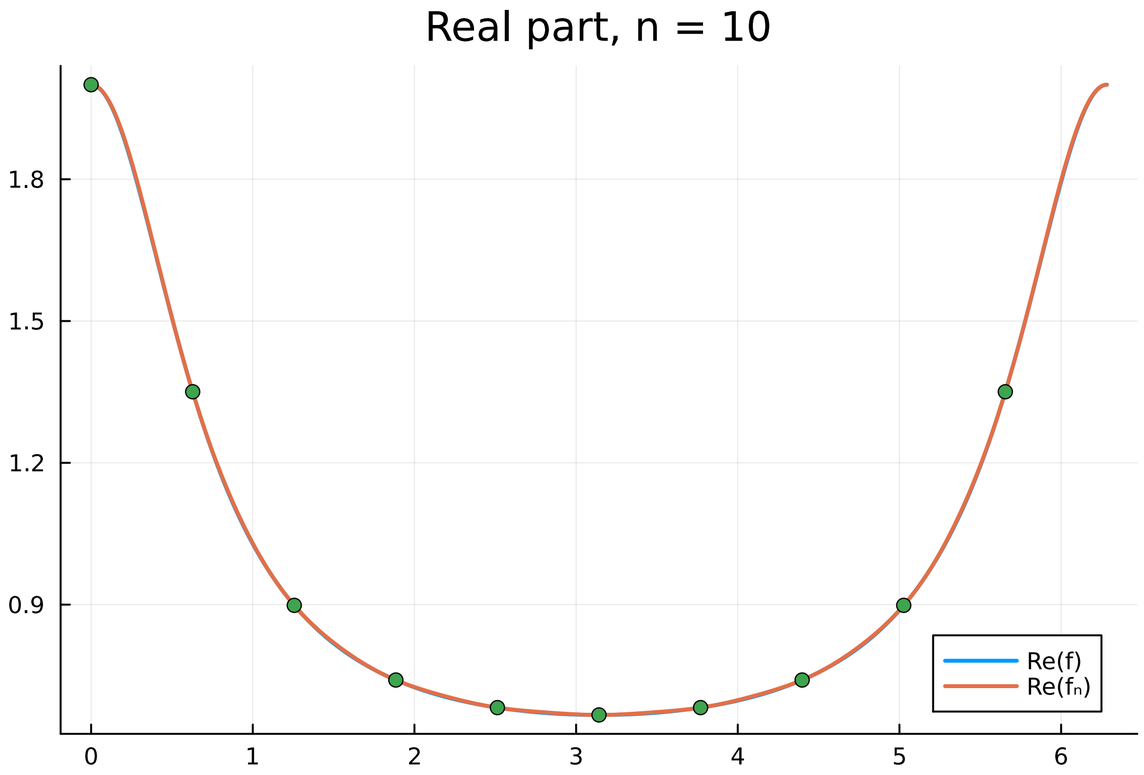
converges uniformly to $f(\theta)$*.*

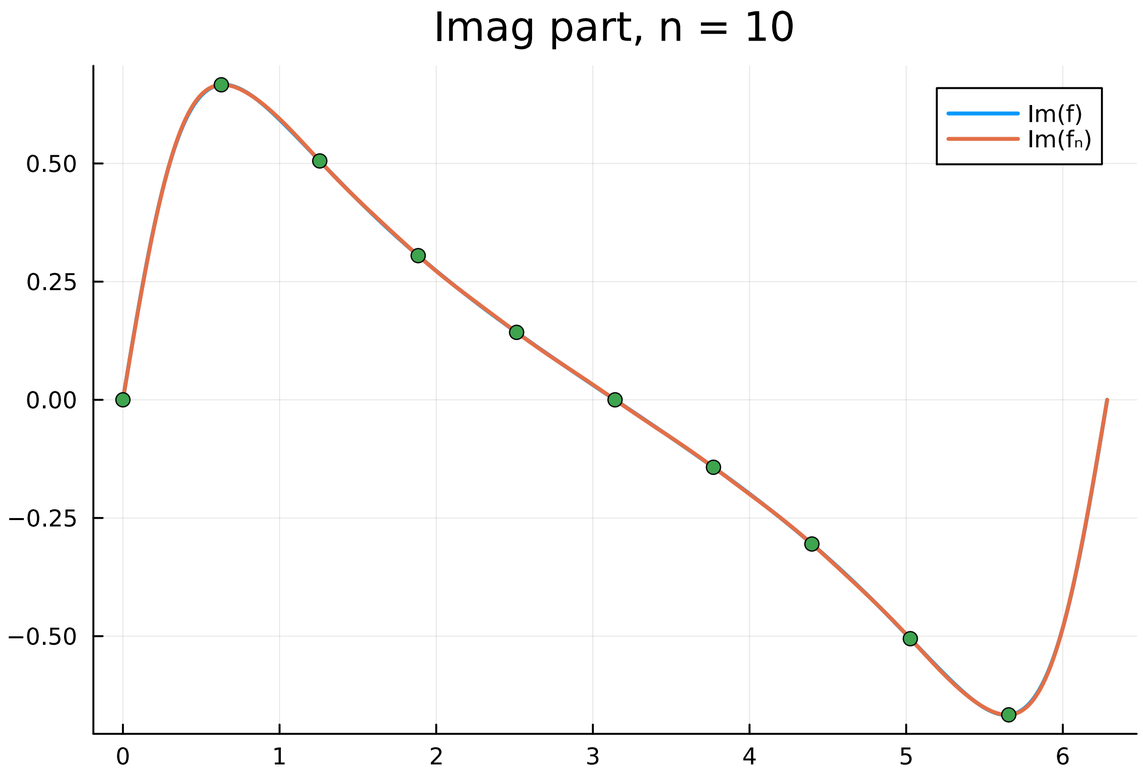
$$f(\theta) = \frac{2}{2 - \mathrm{e}^{\mathrm{i}\theta}}$$



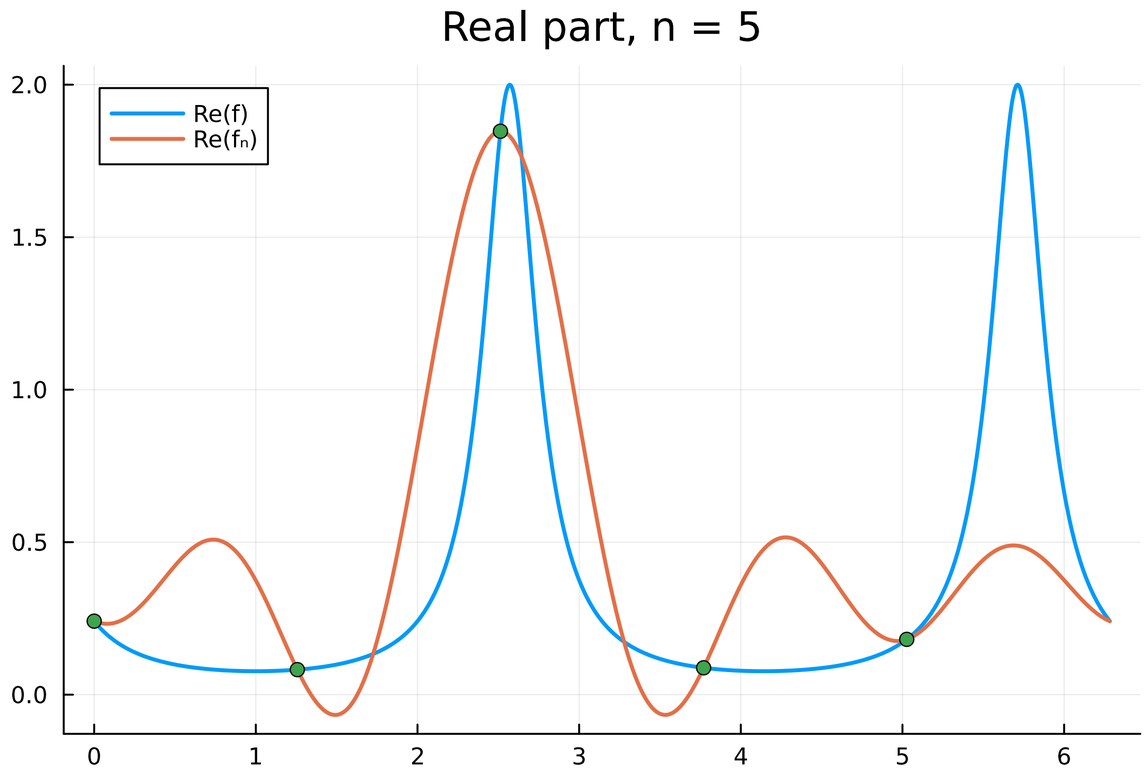


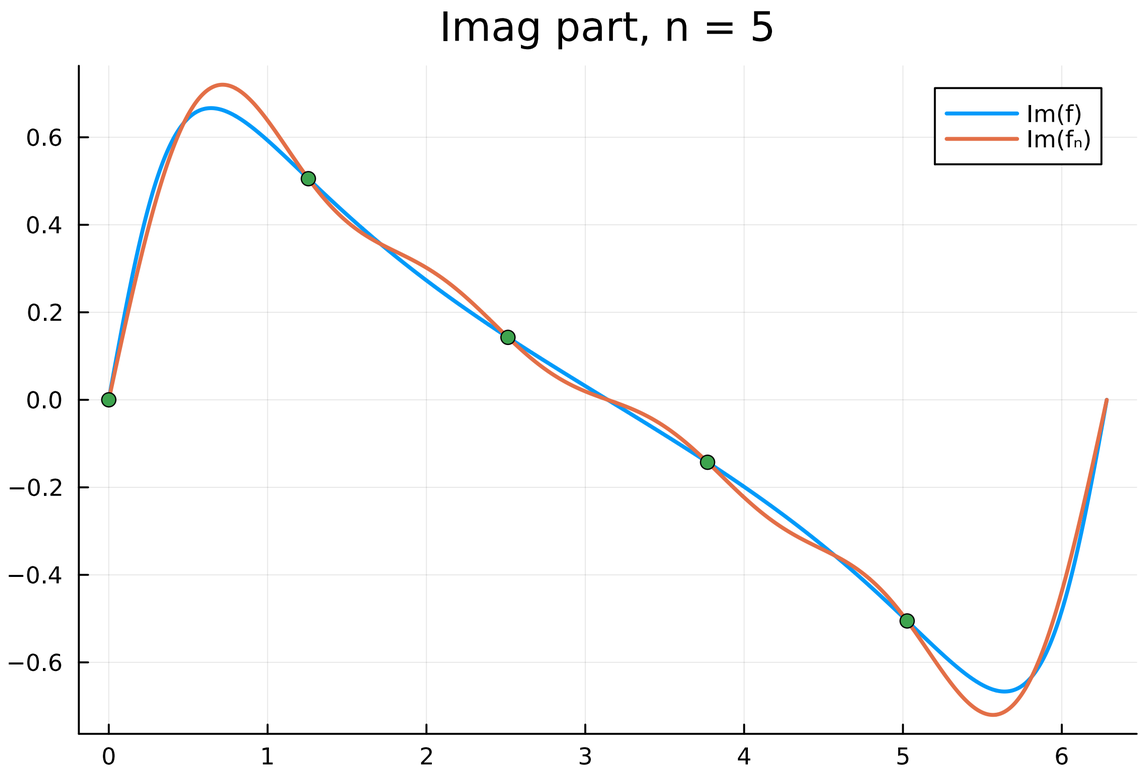
$$f(\theta) = \frac{2}{2 - \mathrm{e}^{\mathrm{i}\theta}}$$



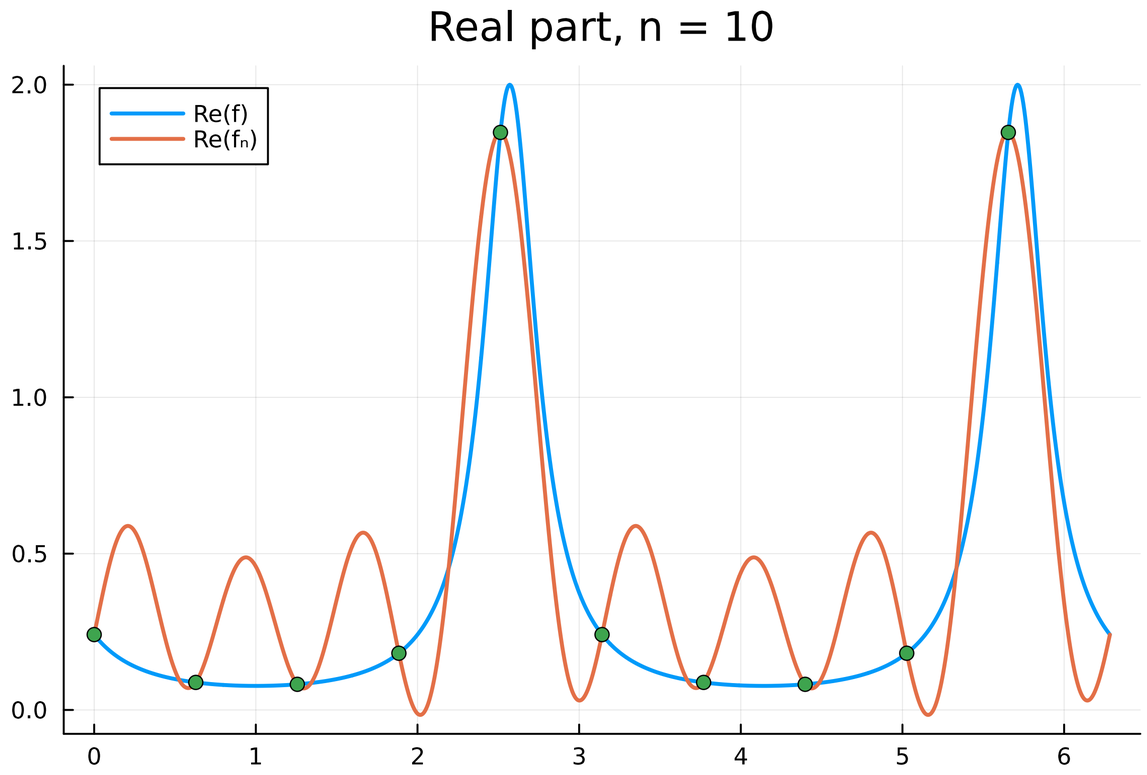


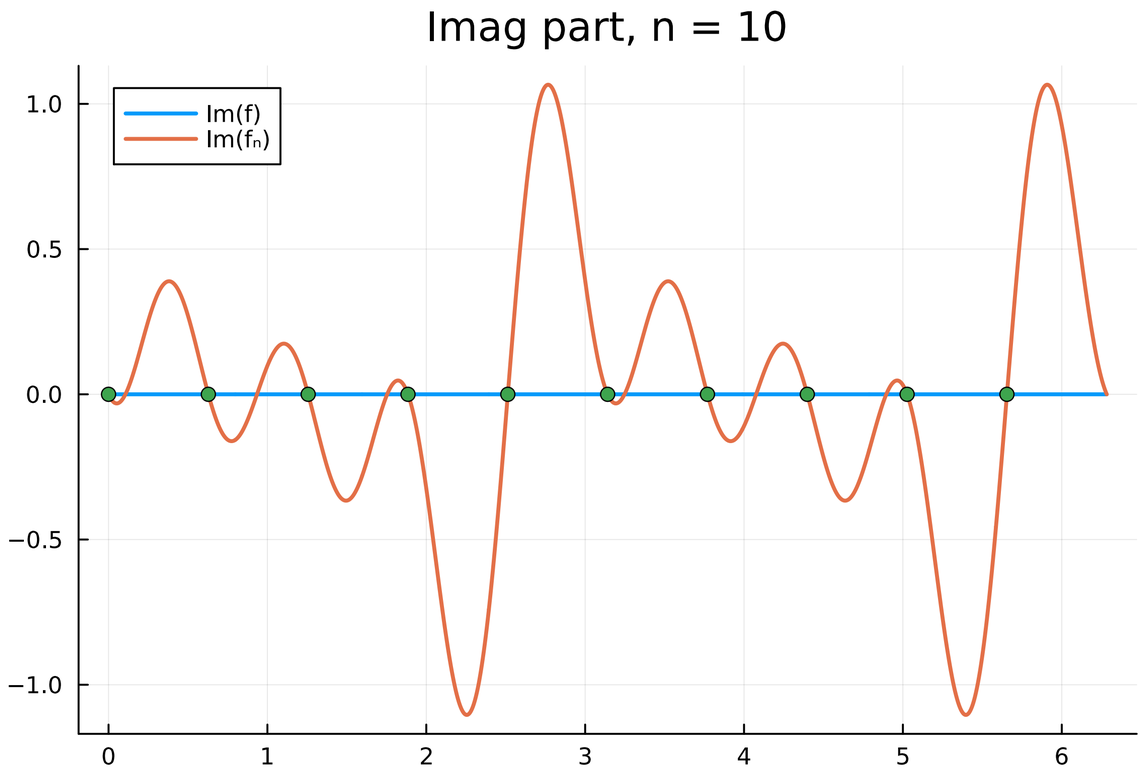
$$f(\theta) = \frac{2}{25\cos(\theta - 1)^2 + 1}$$



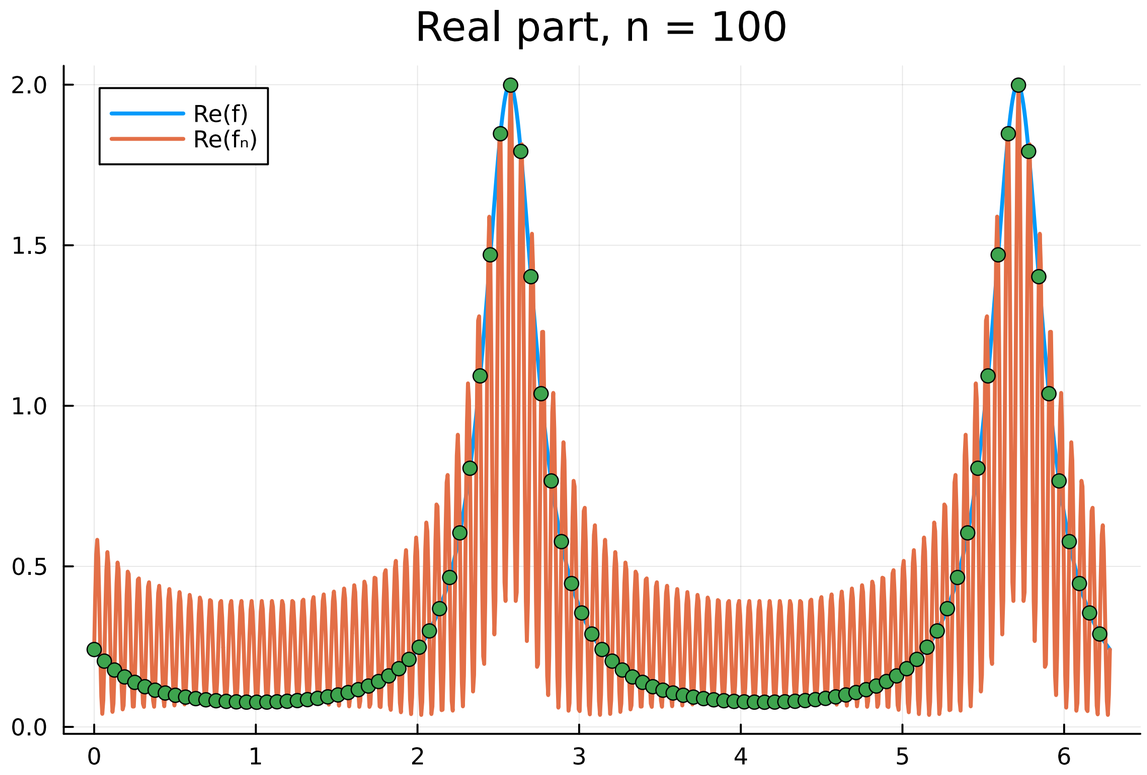


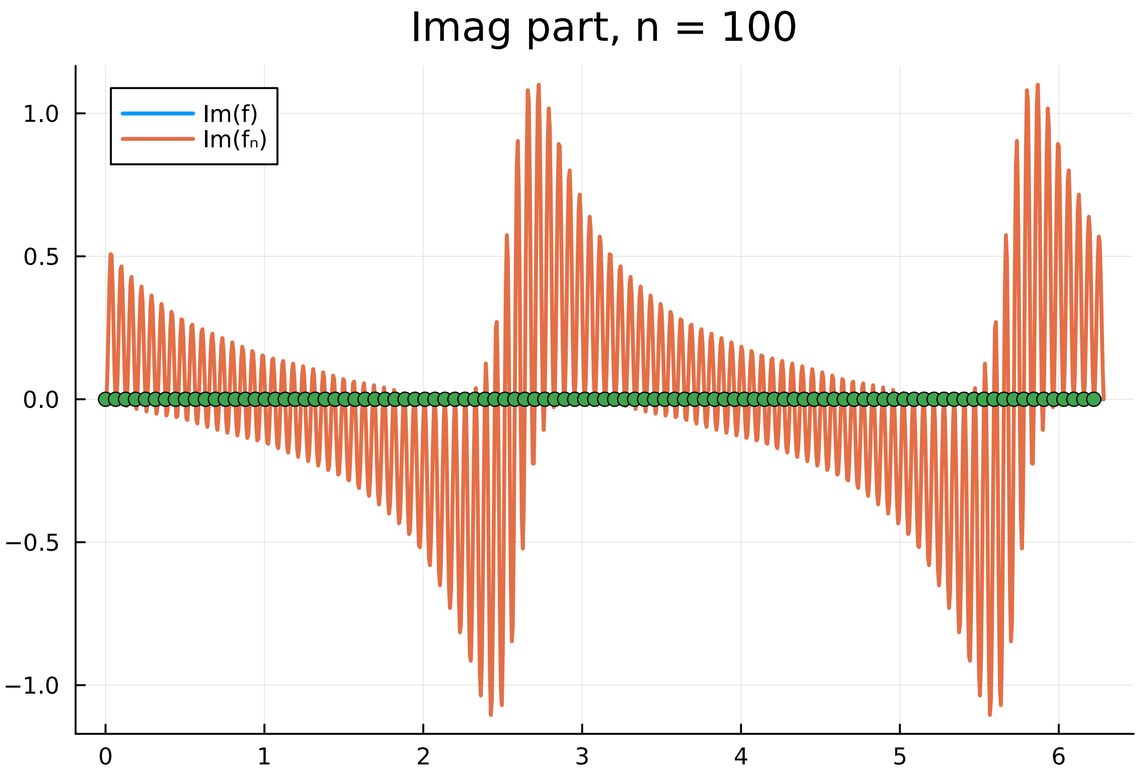
$$f(\theta) = \frac{2}{25\cos(\theta - 1)^2 + 1}$$





$$f(\theta) = \frac{2}{25\cos(\theta - 1)^2 + 1}$$

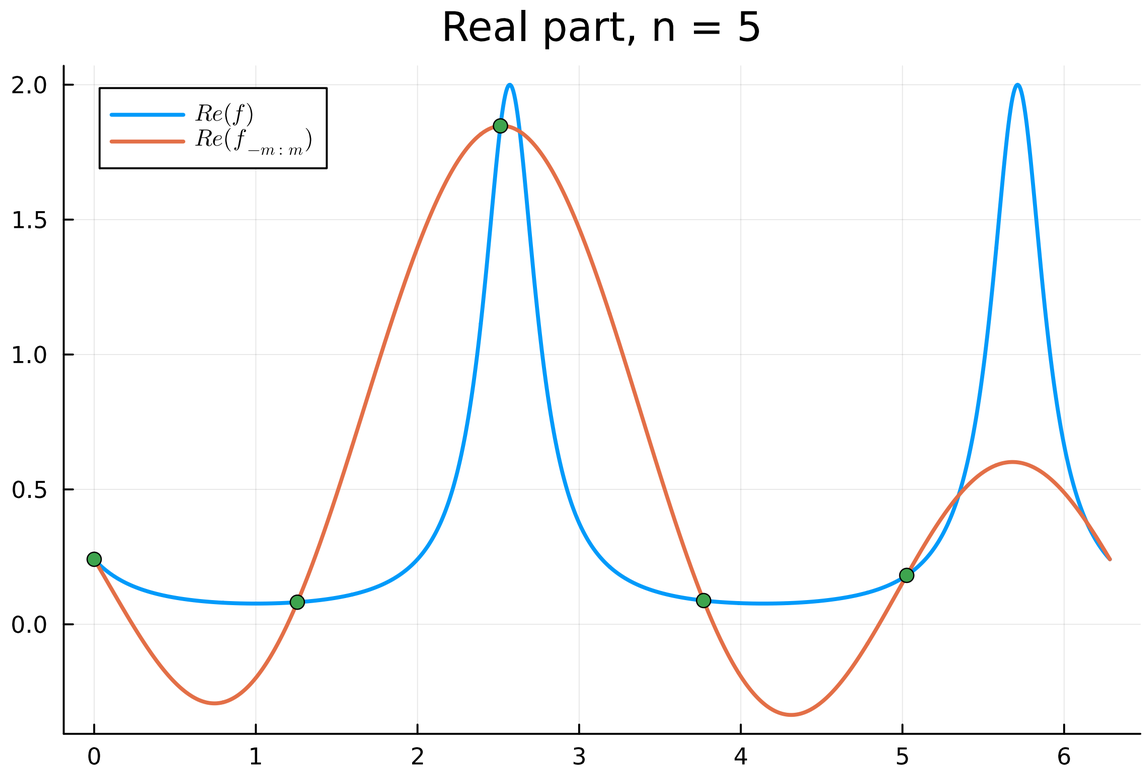


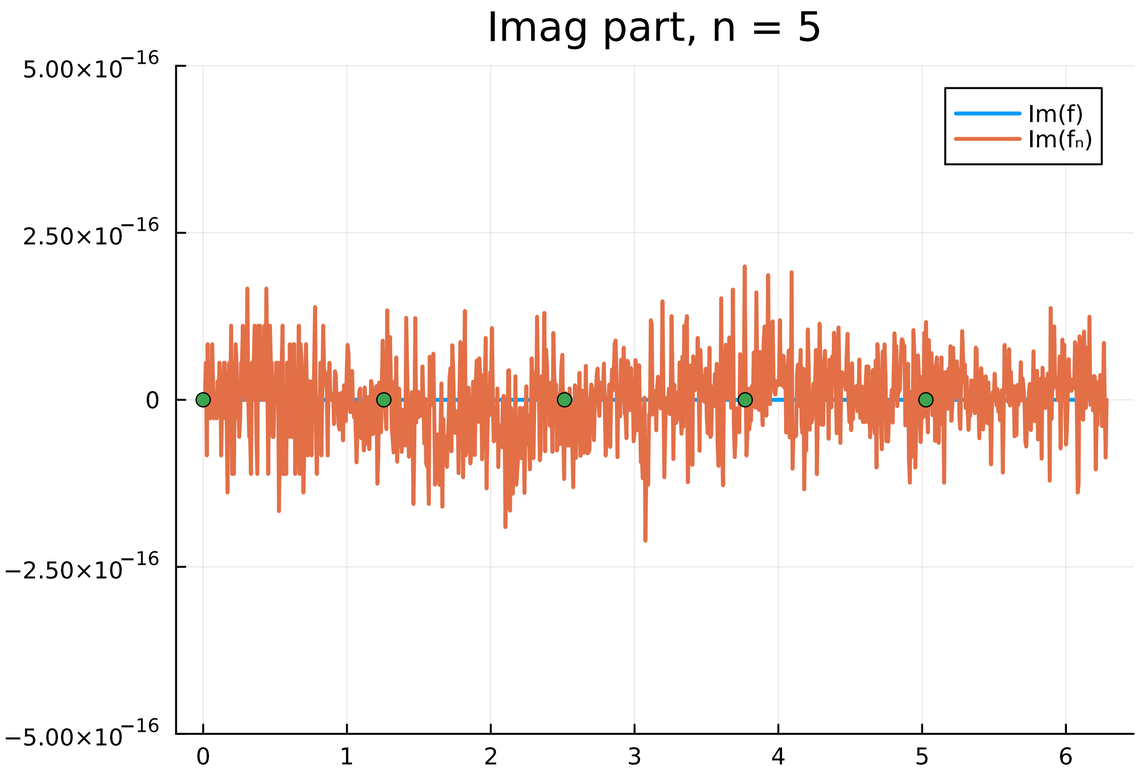


Corollary 3 (aliasing)**.** *For all* $p \in \mathbb{Z}$, $\hat{f}_k^n = \hat{f}_{k+pn}^n$.

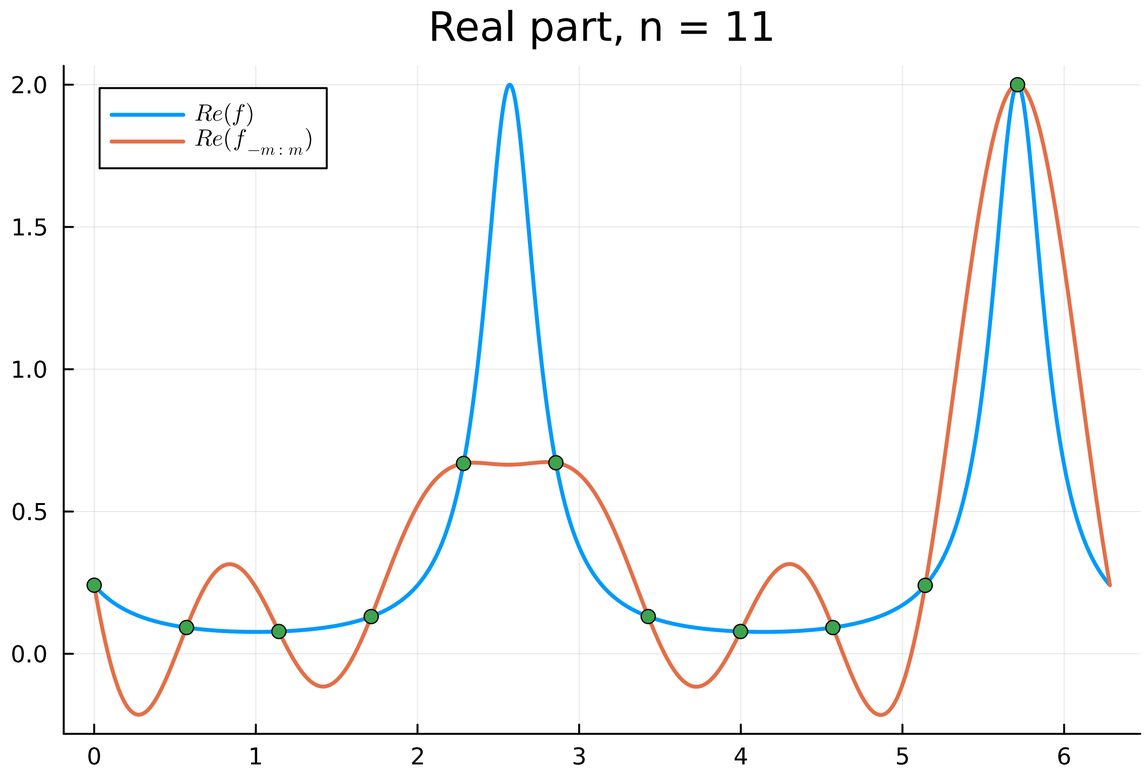
Consider $f_{-m:m}(\theta) := \sum_{k=-m}^{m} \hat{f}_k^n \mathrm{e}^{\mathrm{i}k\theta}$ where $n = 2m + 1$.

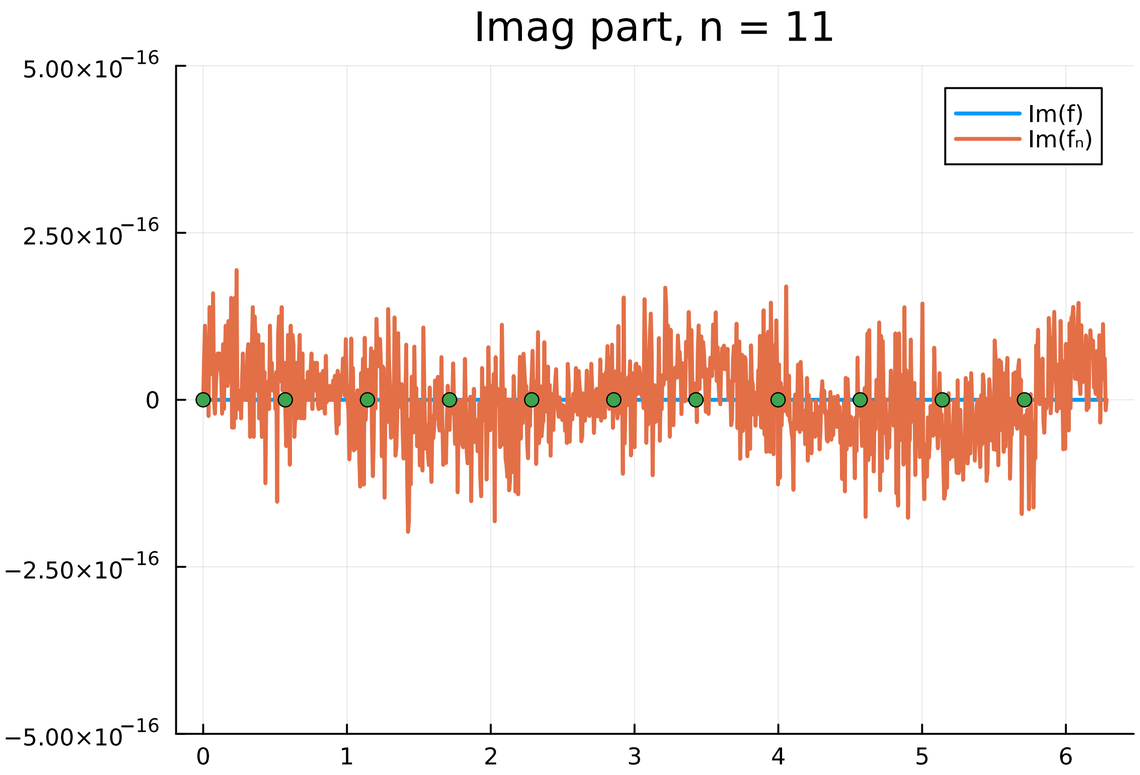
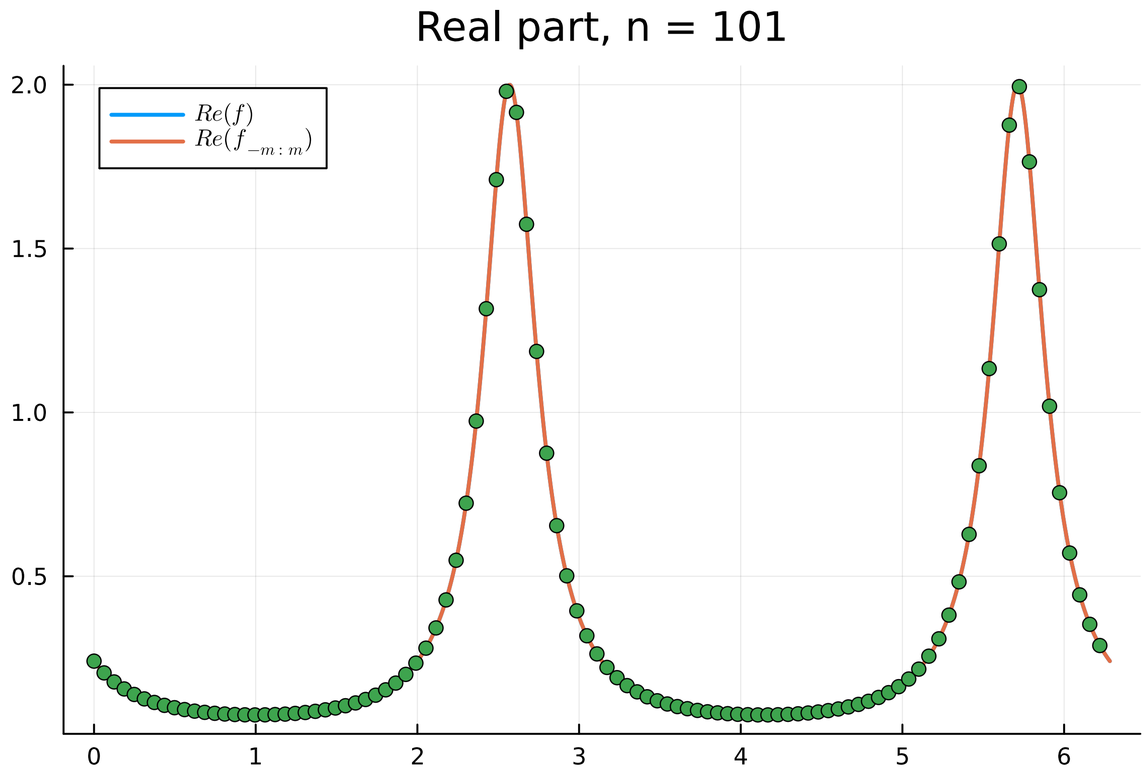
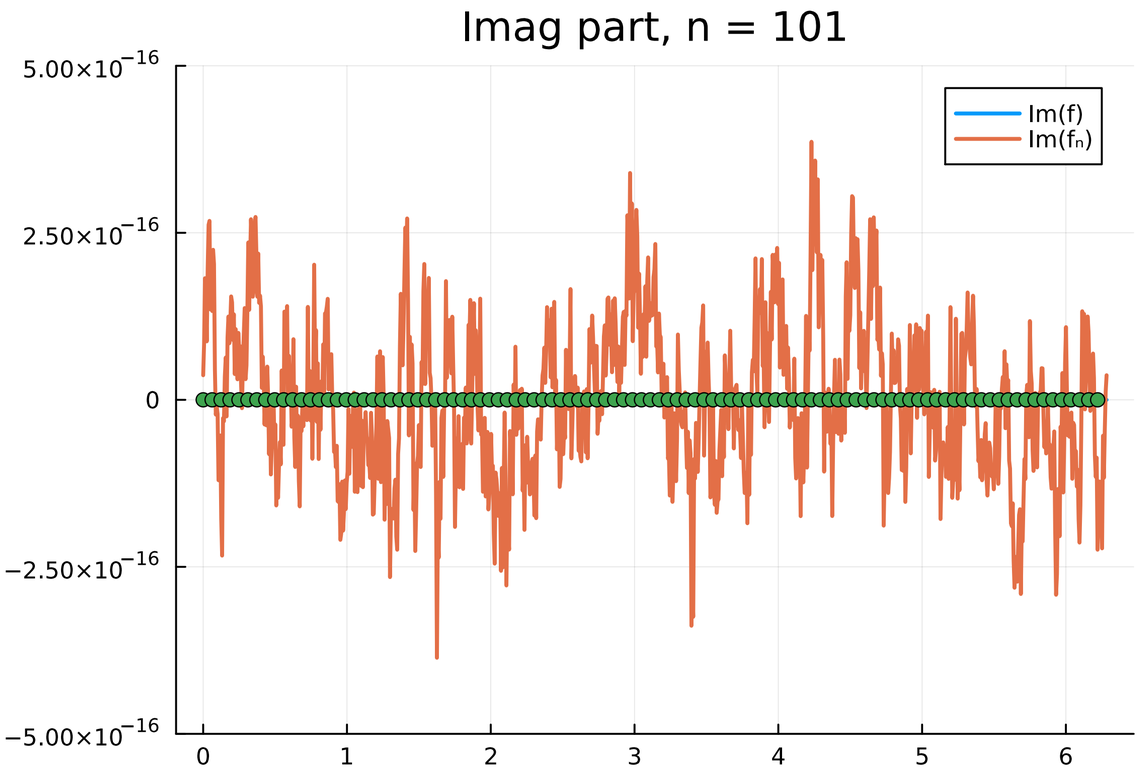
$$f(\theta) = \frac{2}{25\cos(\theta - 1)^2 + 1}$$





$$f(\theta) = \frac{2}{25\cos(\theta - 1)^2 + 1}$$





$$f(\theta) = \frac{2}{25\cos(\theta - 1)^2 + 1}$$